DATA SCIENCE LABORATORY MANUAL

Module II ? Python for Data Science

Topic: NumPy

OVERVIEW & LEARNING OBJECTIVES

N-dimensional arrays, broadcasting, vectorized math, array reshaping, slicing, and linear algebra routines.

KEY TOPIC CONCEPTS & METHODOLOGY

- Understanding NumPy in modern Data Science workflows.
- Detailed breakdown: N-dimensional arrays, broadcasting, vectorized math, array reshaping, slicing, and linear algebra routine
- Mathematical formulas, code examples, and laboratory execution guidelines.
- Common pitfalls, edge cases, and best practices in practical applications.

LABORATORY PRACTICE / IMPLEMENTATION NOTES

1. Follow standard Python and Data Science coding practices.
2. Document all exploratory findings and statistical hypotheses.
3. Keep datasets and outputs organized in your repository workspace.